LeetCode — Maximum Score Good Integer

Language: Java

Question

You are given two positive integers **num** and **sum**. A positive integer is good if it has exactly **num** digits and the sum of its digits is exactly **sum**. Its score is the sum of the squares of its digits. Return the good integer with the maximum score. If multiple integers have the same maximum score, return the largest one. If no such integer exists, return an empty string.

Examples

Example 1
Input: num = 2, sum = 3
Output: "30"
Explanation: 30 has score $3^2 + 0^2 = 9$, which is greater than the scores of 12 and 21.

Example 2
Input: num = 2, sum = 17
Output: "98"
Explanation: Both 89 and 98 have the same score, so choose the larger integer, 98.

Example 3
Input: num = 1, sum = 10
Output: ""
Explanation: One digit cannot have a digit sum of 10.

Java Solution

```
class Solution {
    public String maxSumOfSquares(int num, int sum) {
        if (sum > 9 * num) {
            return "";
        }

        StringBuilder result = new StringBuilder();

        for (int i = 0; i < num; i++) {
            int digit = Math.min(9, sum);
            result.append(digit);
            sum -= digit;
        }

        return result.toString();
    }
}
```

Simple Logic

1. Every digit can be at most 9. If sum > 9 × num, no valid number exists.
2. Put the largest possible digit first: Math.min(9, sum).
3. Subtract that digit from sum, then repeat until num digits are created.
4. Any remaining positions automatically receive 0 once sum becomes 0. Putting larger digits first maximizes the score and gives the largest number in a tie.

Complexity

Time Complexity: O(num)
Space Complexity: O(num), for the output string.